

Tae Hoon Yang

taeyang@stanford.edu — tae-h-yang.github.io — (617) 838-8421

Education

Stanford University, Stanford, CA

Sep 2024 — Expected Jun 2026

M.S. in Computer Science (AI Specialization), GPA: 4.1/4.0

Advisors: Shuran Song, Karen Liu

Northeastern University, Boston, MA

Sep 2018 — May 2024

B.S. in Computer Engineering and Minor in Robotics, Summa Cum Laude

Advisor: Michael Everett

Publications

Locomotion Beyond Feet

Tae Hoon Yang*, Haochen Shi*, Jiacheng Hu*, Zhicong Zhang, Daniel Jiang, Weizhuo Wang, Yao He, Zhen Wu, Yifan Hou, Monroe Kennedy III, Shuran Song[†], C. Karen Liu[†],
Arxiv, 2026

Toward Intelligent Design and Measurement of MEMS Acoustic Wave Resonators

Xingyu Liu, Tianxi Meng, Shuhan Chen, **Tae Hoon Yang**, Jun Zhang, Songbin Gong, Yansong Yang
IEEE Transactions on Microwave Theory and Techniques (TMTT), 2026

Selected Projects

Toddlerbot 2.0: Open-Source ML-Compatible Humanoid Platform

Dec 2024 - Aug 2025

Robotics and Embodied Artificial Intelligence (REAL) Lab and The Movement Lab

Stanford University

- Contributed to the 2.0 release by improving the custom keyframe animation app and adding non-physics motion generating and RL training tools.
- Built an RL training pipeline with custom terrain generation, integrated stereo depth perception using the deep learning model FoundationStereo and added RGB-D rendering and elevation mapping features in simulation.

Go1 DLIOM-Based Autonomous Navigation

Jan 2024 - Aug 2024

Autonomy & Intelligence Lab

Northeastern University

- Configured Unitree Go1 robot dog with 3D LiDAR sensor fixture and ROS2 environment on robot's Jetson board.
- Integrated state estimation from DLIOM (Direct LiDAR Inertial Odometry and Mapping) with ROS2 Nav2.
- Achieved robot navigation with custom nodes that convert velocities from Nav2 to robot's high-level SDK commands.

Teaching Experience

CS229 Machine Learning

Autumn 2025, Winter 2026

Instructors: Moses Charikar, Carlos Guestrin, Andrew Ng, Emily Fox, Sanmi Koyejo

Stanford University

- Assisted with comprehensive course support including homework and exam design, grading, and project guidance.

EECE 2160 Embedded Design: Enabling Robotics

Fall 2022, Spring 2024

Instructors: Emad Aboelea, John Kimani

Northeastern University

- Assisted lab sessions and graded assignments covering FPGA and Digital Logic Design (Quartus Prime), Linux, C++ (pointers, dynamic memory, data structures, and OOP), and robotic arm programming.

EECE 2140 Computing Fundamentals

Spring 2023

Instructor: Roi Yehoshua

Northeastern University

- Assisted and graded assignments covering Python programming syntax, DSA, OOP, NumPy, and unit testing.

Work Experience

Hardware Automation Engineering Intern

Jul 2023 - Dec 2023

iRobot

Bedford, MA

- Developed a vision-based dirtiness grading system to quantitatively evaluate cleaning performance of robot wipers.
- Designed an evaluation system using motion-capture data to assess robotic vacuum navigation performance in terms of cleaning coverage efficiency and path consistency.
- Created power supply C# libraries for automated power supply control by implementing SCPI command interfaces over USB and RS-232 connections.

Skills

Programming: Python, C, C++, C#, MATLAB | **Development Tools:** PyTorch, OpenCV, ROS, ROS2, MuJoCo, Linux, Git, Docker | **Hardware & Electronics:** Stanford Toddlerbot, Unitree Go1, 2D/3D LiDAR, Fisheye/Stereo/RGB-D Camera, IMU, GNSS/RTK, Nvidia Jetson, Raspberry Pi, STM32